ENGINEERING ROADMAP • 2026 – 2028

RoboNova

An Autonomous Personal AI Robot — Master Roadmap & Technical Documentation

DURATION	BUDGET	TRACKS	MILESTONES
2 Years	$500	AI + Robotics	5 Versions

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github.com/24pwai0015-max/robonova

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Executive Summary

RoboNova is a two-year, self-funded engineering programme to design, build, and ship a personal AI robot — combining an LLM-driven “brain” with a physical, sensor-equipped “body.” It is structured as a Final Year Project and a public portfolio piece, documenting every stage from first principles through to a working, presentation-ready robot.

> ***“First learn, then implement.”***
>
> Every topic is studied, understood, and practiced before a single line of implementation code is written. No fundamentals are skipped.

The project runs on two parallel tracks — an AI / automation track that builds RoboNova's decision-making, and a robotics / electronics track that builds its physical form — which converge into a single integrated system by the final milestone.

1 Project Overview

Field	Detail
Project Name	RoboNova
Duration	2 Years
Budget	$500
Deliverable	Final Year Project + Portfolio Project
Repository	github.com/24pwai0015-max/robonova

Vision

A robot that can talk naturally, listen to voice commands, recognize people and objects, move around independently, perform simple chores, remember conversations, and be upgraded over time — built from first principles and documented well enough to stand on its own as an engineering portfolio.

Goal Capabilities

Capability	What it means for RoboNova
Talk naturally	Real-time conversational voice output
Listen to voice commands	Wake-word + speech-to-text pipeline
Recognize people & objects	Computer vision — face & object recognition
Move independently	Motor control + navigation
Perform simple chores	Task planning and execution
Remember conversations	Persistent memory layer
Upgrade over time	Modular hardware & software architecture

2 Guiding Philosophy

Every topic in RoboNova follows the same disciplined sequence — fundamentals are never skipped:

1. Learn
2. Understand
3. Practice
4. Mini Assignment
5. Mini Project

6. Review
7. Move Forward

3 The Two-Track System

RoboNova runs alongside the existing AI / Data Science roadmap (the Agent Automation Developer path) rather than replacing it. The two tracks converge into one integrated AI robot.

Track 1 — Brain (Career/AI)	Track 2 — Body (Robotics)
Python	Electronics
NumPy	Arduino
Pandas	Raspberry Pi
Machine Learning	Sensors
APIs	Motors
Agent Automation	ROS2
LangChain / CrewAI	Robotics Integration

4 Weekly Study Schedule

Day	Activity
Monday	Learn
Tuesday	Learn
Wednesday	Learn
Thursday	Learn
Friday	Rest
Saturday	Learn + Practice
Sunday	Review + Mini Project

Every week closes with a mini project — a small, working artifact that proves the week's concepts.

5 Complete Roadmap — 8 Phases

Phase 0 — Robotics Foundation

Duration: 2 weeks · No hardware yet

- What is robotics? Types of robots. Robot anatomy.
- Sensors, actuators, controllers, power systems, communication
- Arduino, Raspberry Pi, ESP32 — platform landscape
- The Sense → Think → Act loop

Phase 1 — Electronics

- Voltage, current, resistance, Ohm's Law
- Breadboarding — LEDs, buttons, circuits
- Servo motors, DC motors, motor drivers

Phase 2 — Arduino

- LED control · button input · distance sensing
- Servo control · motor control

Phase 3 — Raspberry Pi

- Linux fundamentals · GPIO · Python on-device
- Camera module · networking · SSH

Phase 4 — Computer Vision

- Face detection & recognition
- Object detection & tracking

Phase 5 — ROS2

- Nodes, topics, publishers, subscribers
- Services and launch files

Phase 6 — RoboNova Prototype

- Integrate camera, speaker, microphone, motors, sensors, Pi, and Arduino into one physical unit

Phase 7 — AI Integration

- LLM reasoning · persistent memory · planning
- Voice assistant · task execution

6 Milestones

Progress is tracked through versioned releases rather than a single two-year finish line:

v0.1	Architecture and design
v0.5	Sensors and movement
v1.0	Mobile robot with vision
v2.0	Voice assistant and AI integration
Final	Complete, presentation-ready robot

7 Week 1 — Starting Point

Monday — Introduction to Robotics

- What is robotics? What is a robot? Brief history. Types. Components.

Assignment: one-page explanation of "What is a robot?"

Tuesday — Robot Anatomy

- Sensors, actuators, controller, power, mechanical structure, communication

Assignment: diagram RoboNova and label all six parts.

Wednesday — Robot Brain

- Arduino, ESP32, Raspberry Pi, mini PCs — when to use each

Assignment: build a comparison table.

Thursday — How Robots Think

- Sense, Think, Act, feedback, open-loop vs. closed-loop systems

Assignment: diagram RoboNova's decision flow.

Friday — Rest

Saturday — Real Robots

- Research humanoid, industrial, service, medical, and warehouse robots

Assignment: analyze three robots' sensors, movement, and purpose.

Sunday — Mini Project: Design RoboNova v0.1

- Mission, features, sensors, actuators, brain, power, communication, architecture diagram

No coding this week — design only.

8 Project Organization

Folder Structure

```
RoboNova/
├── docs/
├── learning_notes/
├── hardware/
├── software/
├── simulations/
├── experiments/
├── images/
└── README.md
```

Daily Engineering Log Template

```
Date:
Today's Topic:
Key Concepts:
Questions:
How It Helps RoboNova:
Time Spent:
```

9 Teaching Method

Every lesson follows this structure — closer to a university engineering course than a collection of tutorials:

8. Theory
9. Why it matters
10. How RoboNova will use it
11. Real-world examples
12. Homework
13. Short quiz
14. Weekly mini project

10 Where to Start

Begin at Week 1 — Monday: "Introduction to Robotics." Today is Day 1. There is no catch-up penalty — the priority is a solid, well-documented foundation, built one step at a time.